

# Boltzmann Brain: A Versioned, Distributable, and Model-Agnostic Knowledge Architecture

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## Abstract

Knowledge produced with large language models (LLMs) tends to become locked inside a single provider, agent, or vector store, where it is hard to audit, reindex, version, or move. We propose **Boltzmann Brain**, a portable knowledge architecture built on one principle: the brain conserves, validates, and retrieves knowledge, while an external and interchangeable LLM processes, contextualizes, and uses it. The brain never embeds a model of its own. Knowledge is stored as typed, content-addressed blocks organized into five memory modules (canonical, episodic, semantic, procedural, and provenance), inspired by complementary learning systems and hippocampal indexing rather than by literal brain-to-database mappings. Each module bundles its blocks, a Merkle DAG that pins the exact composition of a version, and the indices needed to query it. We separate three levels of hashes, one for block identity, one for the Merkle root of a snapshot, and one for the OCI digest of a distributed blob, and show how packaging each module as a separate OCI Artifact yields selective installation and incremental updates. Interaction happens through the *Boltzmann Protocol*: during ingestion and query the LLM proposes candidate blocks and consumes Evidence Bundles, while the protocol governs what is committed and returns only verified evidence, never prose. Because the brain is portable data addressed by a protocol rather than a deployable service, the same snapshot can be read and extended by any conforming client. Canonical blocks are the root of evidence: they record what was observed, not what is true, and every derived interpretation cites them through provenance. Because wrong knowledge must be removable without punching holes in the current snapshot, we specify how non-episodic modules drop blocks, rebuild their Merkle DAGs, and cascade through provenance, with a privileged cascade when the dropped block is canonical, and with redaction reserved for destroying retained bytes. This first version fixes the boundaries and responsibilities of the architecture and leaves concrete block schemas, wire formats, and algorithms for later validation.

**Keywords:** knowledge representation, retrieval-augmented generation, content addressing, Merkle DAG, OCI Artifacts, memory systems, LLM agents

## 1 Introduction

Large language models have made it cheap to *produce* knowledge (notes, summaries, extracted facts, procedures), but expensive to *own* it. In practice, the knowledge a user accumulates while working with an LLM tends to be captured in a form dictated by whichever provider, agent framework, or vector database happens to be in use. Embeddings are computed with a specific

model, chunks are shaped by a specific pipeline, and the resulting store is rarely portable, verifiable, or re-interpretable. When the model changes, the provider changes, or the indexing strategy changes, the knowledge is often effectively lost, or must be rebuilt from scratch.

This paper argues that knowledge should be a *first-class, portable artifact* that outlives any particular model or vendor. We present **Boltzmann Brain**, an architecture whose sole responsibility is to conserve information, organize it into memories with distinct purposes, maintain verifiable versions, and retrieve relevant evidence. Crucially, the brain contains no language model of its own. It integrates with whatever LLM the user chooses, which interprets sources during ingestion and contextualizes results during query. The design is captured by a single guiding idea:

*The brain conserves, validates, and retrieves knowledge. An external LLM processes, contextualizes, and uses it.*

**The responsibility boundary.** The brain does not write the final answer. It returns knowledge blocks and evidence; an agent, interface, or downstream application decides how to contextualize them (Figure 1). This boundary is what keeps the architecture independent of any provider: the Boltzmann Protocol governs structure, identity, validation, and distribution, while the semantic work (reading, extracting concepts, recognizing episodes, proposing procedures, and phrasing answers) is delegated to an interchangeable external model.

**Contributions.** This work makes the following contributions:

1. A set of design principles and problems (Section 2) for portable, verifiable, model-agnostic knowledge, and an argument for why six requirements that existing work satisfies separately must hold over a single artifact (Section 4).
2. A mapping from biological memory principles and classic systems structures to concrete architectural decisions (Section 3).
3. A five-module memory model (canonical, episodic, semantic, procedural, and provenance; Section 5) that treats the canonical module as the root of observed evidence, together with an internal module anatomy of content-addressed blocks, Merkle DAGs, and indices (Section 6).
4. A distribution model based on OCI Artifacts that enables selective installation and incremental updates (Section 7), and a model-agnostic protocol with explicit ingestion and query contracts (Sections 8–10).
5. A treatment of how knowledge leaves a brain (Section 11): drop with Merkle rebuild and provenance cascade for non-episodic modules, a privileged cascade when canonical evidence is excluded, supersession for episodic memory, pruning of unreachable blocks, and redaction when retained bytes must be destroyed.

**Scope of this version.** This document fixes an initial conceptual architecture. It deliberately does not close the exact schema of each block, the consolidation algorithm, the size of packs, or the concrete index engine. The goal is to establish boundaries and responsibilities that later, more concrete work can validate.

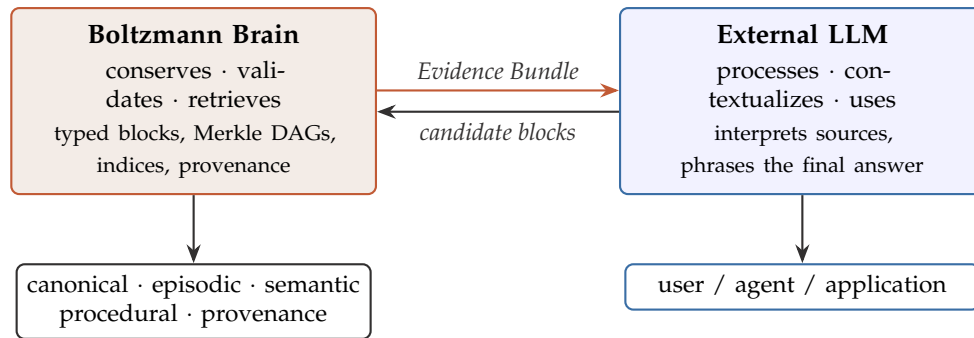


Figure 1: Separation of responsibilities. The brain owns knowledge, identity, and verification; the external LLM owns interpretation and phrasing. The two communicate through a stable contract: the LLM sends candidate blocks during ingestion and receives Evidence Bundles during query.

## 2 Design Principles and Problems

### 2.1 Problems this architecture addresses

Boltzmann Brain is designed to address a cluster of related failures in how LLM-era knowledge is currently stored and consumed:

- Preventing knowledge from being trapped inside a particular provider, agent, or vector database.
- Maintaining a source of evidence that allows knowledge to be reindexed and reinterpreted as models and methods evolve.
- Distinguishing concrete experiences, general facts, and procedures, rather than collapsing everything into undifferentiated chunks.
- Updating knowledge without losing history, provenance, or reproducibility.
- Removing knowledge that is obsolete, costly, or legally untenable, without silently compromising verifiability.
- Distributing a complete brain or only selected modules.
- Allowing interchangeable external LLMs to query and contribute knowledge through a stable contract.

### 2.2 Guiding principles

Eight principles follow from these problems and constrain every subsequent design decision:

1. Knowledge lives in blocks; indices are not the source of truth.
2. Original evidence is preserved by default; interpretations may change. Dropping a canonical block is allowed only under explicit policy, and it forfeits re-derivation from that source.
3. Every version must be identifiable, verifiable, and reconstructible.
4. Modules must be distributable and installable selectively.
5. The protocol must not embed an LLM of its own nor depend on a specific provider.

6. Specialized structures compose; no single index answers every query.
7. Queries are declarative and index-agnostic: callers ask for knowledge, and the selection and combination of indices is left to the implementation.
8. Forgetting is explicit and auditable: non-episodic modules support drop with Merkle rebuild and provenance cascade; dropping canonical evidence is privileged and always cascades; content may be redacted under policy, but never silently.

These principles are intentionally conservative. They privilege durability and auditability over convenience, and they treat the LLM as a powerful but replaceable interpreter rather than as the system of record.

## 3 Inspirations

### 3.1 Biology of memory

The architecture takes *general principles* from neuroscience, not literal equivalences between brain regions and databases. The scientific literature describes multiple memory systems and an interaction among fast learning, stable knowledge, consolidation, associative recall, and adaptive forgetting [8, 11, 12, 15, 20–22, 26]. Figure 2 summarizes the translation from biological principles to architectural decisions.

Complementary learning systems (CLS) theory distinguishes a fast, hippocampus-associated mechanism from slower, integrative cortical learning [12]. For Boltzmann Brain this suggests separating the *capture of episodes* from *semantic consolidation*. The hippocampal indexing theory further motivates a distinction between the index that reactivates a memory and the distributed content of that memory [8], a distinction we mirror in our separation of indices from knowledge blocks. Pattern separation inspires deduplication, identity resolution, and contradiction detection; pattern completion inspires recovering a full memory from a partial query through hybrid search [11, 20, 26]. Finally, reconsolidation suggests that retrieving knowledge can open a controlled revision [15, 21], while work on pruning and active forgetting suggests that removing what is not reinforced is a regulated process rather than a failure, and that accessibility can be reduced without destroying the underlying evidence [6, 19, 22]. Section 11 develops this last point into concrete retention and erasure mechanisms.

### 3.2 Systems structures

Alongside biology, the architecture reuses well-understood systems structures. Table 1 lists each one and its contribution.

Table 1: Systems structures and their contribution to Boltzmann Brain.

| Concept                   | Contribution to Boltzmann Brain                        |
|---------------------------|--------------------------------------------------------|
| Content addressing        | The identity of a block derives from its content.      |
| Merkle DAG                | A root identifies the exact composition of a module.   |
| B <sup>+</sup> tree       | Exact lookup, hierarchies, and ordered range scans.    |
| Inverted index            | Lexical search over words and terms.                   |
| Vector index              | Approximate retrieval by semantic similarity.          |
| Graph                     | Explicit relations, provenance, associative expansion. |
| Mark-and-sweep collection | Reclaiming blocks unreachable from a retained root.    |
| OCI Artifact              | Interoperable packaging and distribution of modules.   |
| Skill / MCP / CLI / API   | Contract for an external LLM to query and ingest.      |

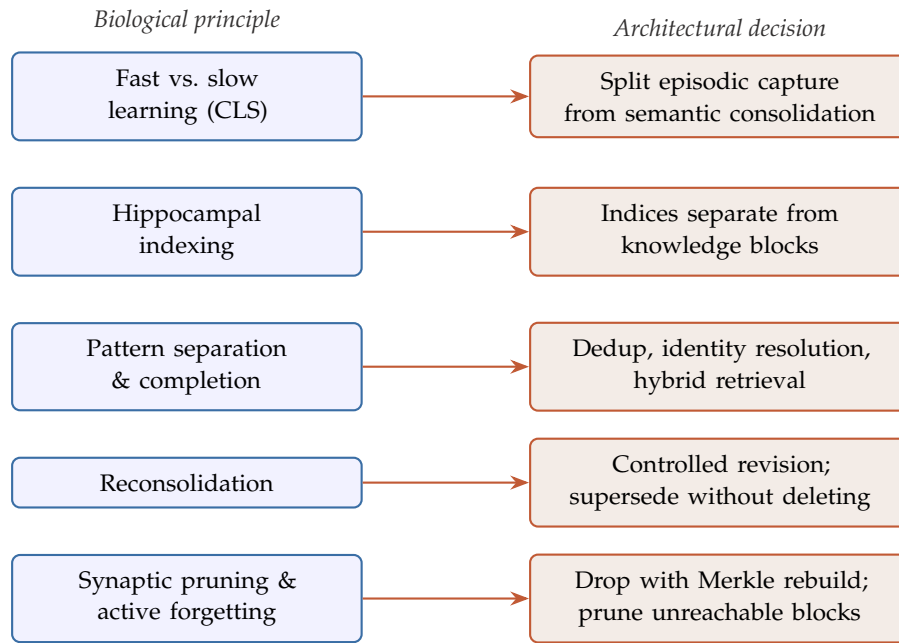


Figure 2: Translation of biological memory principles into architectural decisions. The mapping is by analogy, not by literal correspondence between brain regions and system components.

## 4 Related Work and the Case for a New Protocol

### 4.1 The landscape

The industry is already solving pieces of this problem, but no single artifact yet integrates a canonical source, typed memories, fine-grained versioning, portable indices, modular distribution, and a model-agnostic access protocol. Table 2 positions the most relevant efforts; we then discuss them in turn.

The Open Knowledge Format (OKF) formalizes the knowledge-wiki pattern through Markdown and YAML, emphasizing independence between producers and consumers [9]. GBrain separates durable knowledge from agent behavior and offers interchangeable engines and agent integration [23–25]. GraphRAG and Cogneer illustrate the convergence toward hybrid representations that mix text, vectors, and graphs [4, 5, 13, 14].

**Substrate rather than alternatives.** Two standards that recur throughout this paper are deliberately absent from the comparison. MCP standardizes how applications connect to external resources and tools [1], and the OCI ecosystem provides content-addressed manifests, descriptors, and blobs whose structure is itself an external Merkle DAG [16–18]. Neither is an alternative to a knowledge architecture, because one standardizes a call and the other a transport, and this design adopts both instead of competing with them (Sections 7 and 8). They return in Section 4.3, where the question is what a stack assembled out of them still fails to provide.

### 4.2 Six requirements that must hold simultaneously

Proposing a new protocol requires more than observing that existing tools are imperfect. The argument of this paper is that the problems of Section 2 translate into six requirements which, to be useful, must hold *at the same time and over the same artifact*:

Table 2: Related efforts and their distance from Boltzmann Brain.

| Project / standard    | What it contributes                                                                  | Distance from Boltzmann Brain                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Open Knowledge Format | Markdown + YAML as a portable, human-readable format for agents.                     | Formalizes canonical exchange, but defines no access protocol, biological memory model, indices, or compiled distribution. |
| GBrain                | Git repository as a durable registry; hybrid search, graph, and agent access.        | Reinforces the independence of knowledge base from agent; its primary distribution follows the Git/repository model.       |
| Microsoft GraphRAG    | Extracts entities, relations, communities, and summaries; combines graph and text.   | An indexing and retrieval pipeline, not a distributable format for a complete brain.                                       |
| Cognee                | Turns data into chunks, embeddings, nodes, and relations; combines vector and graph. | Provides memory operations and enrichment, but does not define the proposed OCI/Merkle bundle.                             |

- R1. Canonical evidence.** The original source is preserved and addressable, so that any interpretation can be audited against it and re-derived when methods improve. Without this, knowledge cannot be re-interpreted, only trusted.
- R2. Typed memories.** Concrete experiences, consolidated facts, and procedures are distinguished rather than collapsed into undifferentiated chunks. Without this, “what happened in the class of May 14” and “the definition of a Fourier series” compete in a single similarity ranking.
- R3. Verifiable versioning at knowledge granularity.** Versions are identifiable, verifiable, and reconstructible at the granularity of a unit of knowledge rather than a file. Without this, a corrected fact cannot be distinguished from an edited document.
- R4. Derived, replaceable indices.** Indices are views that can be rebuilt or replaced when an engine or embedding model changes. Without this, knowledge dies with the model that indexed it.
- R5. Selective and incremental distribution.** A consumer can install one module and update only what changed. Without this, sharing knowledge means shipping everything, every time.
- R6. Model-agnostic access contract.** A stable, declarative contract lets interchangeable models query and contribute. Without this, the store is reachable only through whoever wrote the integration.

Table 3 scores the efforts of Table 2 against these six axes. The pattern is not that existing work is weak; it is that each effort is strong on a different axis. Portable formats concentrate on R1 and leave retrieval undefined, while retrieval pipelines concentrate on R4 and treat versioning and distribution as someone else’s problem. The closest neighbor, GBrain, is partial on almost every axis precisely because it inherits its versioning and distribution from Git rather

than defining them at the knowledge layer. What no row does is combine more than a fraction of the six, which is the gap this paper addresses.

Table 3: Design coverage of the six requirements of Section 4.2: ● full, ◐ partial, ○ absent. The last row states the design intent of this version, not a validated implementation.

| Project / standard          | R1 | R2 | R3 | R4 | R5 | R6 |
|-----------------------------|----|----|----|----|----|----|
| Open Knowledge Format       | ●  | ◐  | ◐  | ○  | ○  | ○  |
| GBrain                      | ●  | ◐  | ◐  | ◐  | ◐  | ◐  |
| Microsoft GraphRAG          | ◐  | ◐  | ○  | ◐  | ○  | ○  |
| Cognee                      | ◐  | ◐  | ○  | ◐  | ○  | ◐  |
| Vector store + RAG pipeline | ○  | ○  | ○  | ○  | ○  | ○  |
| Boltzmann Brain (this work) | ●  | ●  | ●  | ●  | ●  | ●  |

### 4.3 Why composition is not enough

Every one of the six requirements is met somewhere in the wider ecosystem, once the standards of the previous subsection are counted, so the obvious objection is that the answer is integration rather than a new protocol: put the sources in Git, run GraphRAG over them, expose the result through MCP, and ship it with OCI. That stack is worth building, and much of it is what a conforming implementation would look like. What it does not produce is a portable artifact, for four reasons.

**Identity does not compose.** Each layer carries its own notion of what a thing is: a commit, a pipeline output, a registry digest. None of them binds a retrieved item to the source it came from and to the snapshot it belongs to. That binding is exactly what an Evidence Bundle asserts when it reports verified, and it is why this architecture separates three levels of hashes (Section 6) instead of borrowing one. Verifiability is a property of the whole path from source to answer, so it cannot be added by a layer that only sees its own inputs.

**Granularity mismatch.** Version control versions files; knowledge changes at the granularity of a claim. A corrected definition, its evidence, and the procedures that depended on it are not a diff over lines, and recovering that structure from a file history means reconstructing after the fact what the ingestion step already knew. R3 asks for versioning where the knowledge is, which is the block.

**The layering runs the wrong way.** In a retrieval pipeline the index is the product: documents are consumed to build embeddings and graphs, and those derived structures become the thing that is queried, stored, and eventually migrated. R4 requires the opposite ordering, with blocks as the durable artifact and indices as disposable views over them. Inverting an existing pipeline is not a matter of configuration, because a pipeline whose output is its state has no representation of the knowledge that survives it.

**Access and transport standardize the edges, not the middle.** MCP standardizes the call and OCI standardizes the bytes, which is why both appear in this design rather than being replaced by it. Neither says what is on the other side of the call or inside the blob. Two MCP servers over the same corpus share no verifiable state, and two registries holding the same brain agree on digests while knowing nothing about modules, memory types, or snapshots. The missing piece



sits between them: a format for typed, content-addressed blocks plus a contract for reading and extending them.

**Why now.** The same argument would have been premature a few years ago, when the model, the pipeline, and the application were usually chosen once. Today embedding models are deprecated on a timescale of months and agent frameworks turn over faster still, so the useful life of accumulated knowledge routinely exceeds the life of the tools that produced it. The container ecosystem went through this transition: engines and registries became interchangeable only once image identity and packaging were specified independently of any implementation. LLM knowledge is at the earlier stage, with capable engines and no shared artifact, and this paper argues for specifying the artifact.

**Complementarity.** Boltzmann Brain is therefore complementary rather than competing. It can consume OKF-style canonical sources, borrow hybrid retrieval ideas from GraphRAG and Cognee, expose an MCP interface, and ride on OCI for distribution, while contributing the missing piece: a versioned, typed, model-agnostic *bundle* whose modules can be installed and updated selectively, and whose contents remain verifiable no matter which model reads them.

## 5 Knowledge Modules

The brain’s OCI Artifact can contain five logical modules. Each module is installable as an independent physical unit and contains blocks, a Merkle DAG, and the indices needed for query. Table 4 names each module by the question it answers.

Table 4: The five memory modules, illustrated with a university brain.

| Module     | Question it answers                                | Examples                                                            |
|------------|----------------------------------------------------|---------------------------------------------------------------------|
| Canonical  | What material was observed?                        | Notes, books, PDFs, lectures, code, datasets.                       |
| Episodic   | What happened in a concrete context?               | Class of May 14, an error in an exercise, a professor’s correction. |
| Semantic   | What general knowledge was consolidated?           | Definitions, formulas, facts, relations.                            |
| Procedural | How is a task performed?                           | Solving an integral, computing Fourier coefficients.                |
| Provenance | Where did it come from and how was it transformed? | Page, source, pipeline, model, validator, date.                     |

**Canonical module.** Preserves observed evidence, not truth claims. A canonical block asserts that a source was incorporated and that it *contains* certain bytes; it does not declare those bytes correct. Because every semantic and procedural interpretation cites canonical evidence through provenance, this module is the root of re-derivation: without the source, derived knowledge cannot be rebuilt cleanly from below.

The module holds two content-addressed layers. The *original* is the blob as observed (PDF, repository archive, conversation export, image, audio, and so on). An optional *normalized view* is a deterministic transform of that blob (plain text, Markdown, or a structured extract) produced by a named pipeline whose identity is recorded in provenance. Normalized views live in canonical rather than semantic memory because they are still evidence for ingestion,



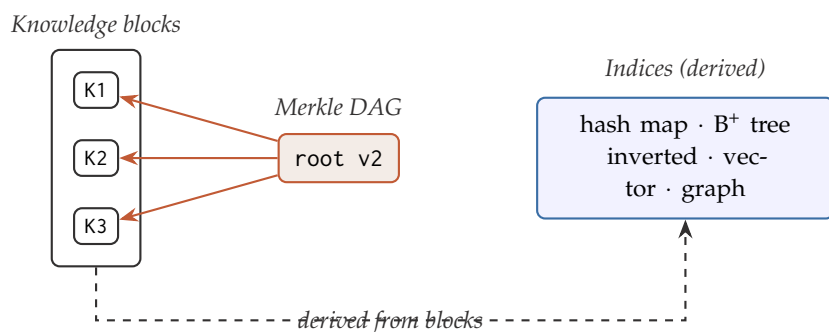


Figure 3: Blocks, a Merkle DAG, and indices coexist within the same module. The Merkle root pins the exact set of blocks that make up a version; indices are derived and can be rebuilt from the blocks.

not consolidated knowledge. Both layers are addressed by hash, so re-registering an identical original is a no-op.

A canonical block therefore carries only what its bytes determine: its `block_id`, media type, size, and optionally the digest of its normalized view. It carries nothing else, because its identity must depend on the observed bytes alone for re-registering an identical original to be a no-op: if the registering actor or the moment of registration were part of the hashed content, two people incorporating the same source would obtain two different blocks and deduplication would never trigger. Everything historical or actor-dependent is recorded in the provenance module instead, which is where relations that depend on who acted and when already live: the license or retention policy under which the source is held, the actor who incorporated it, the timestamp, and the supersession edge toward an earlier edition when a new one replaces it (Sections 9 and 11).

**Episodic module.** Records events and experiences with time, context, participants, outcome, and evidence. It is the only module that stays append-only: corrections generate new episodes or supersession relations rather than drops that rewrite the past (Section 11).

**Semantic module.** Holds consolidated general knowledge: concepts, formulas, facts, relations, and constraints. A Fourier-series formula belongs primarily here, linked to its canonical sources.

**Procedural module.** Represents action sequences, conditions, decisions, alternative paths, and success criteria, for example how to obtain the Fourier coefficients of a given function.

**Provenance module.** Connects derived blocks to evidence, transformations, and tool versions. It is key to answering what must be recomputed when a source changes, and it is the ledger of removals: every drop, cascade, and redaction is recorded there so that exclusion stays auditable (Section 11).

## 6 Anatomy of a Module

A module carries three responsibilities that coexist in the same physical unit (Figure 3): *blocks* contain knowledge; the *Merkle DAG* defines which blocks form a version; *indices* make it possible to find blocks without scanning all of them.

## 6.1 Knowledge blocks

A block is a logical unit of knowledge with a deterministic serialization. It may be an episode, a concept, a fact, a relation, a procedure, or a canonical fragment. Its identifier is computed from its content:

$$\text{block\_id} = \text{SHA-256}(\text{canonical\_serialization}(\text{block})). \quad (1)$$

If the content changes, a new block is born. The previous one may be kept for history, audit, or rollback.

## 6.2 Merkle DAG

On their own, blocks are just a bag of content-addressed units: each is addressable by its hash, but nothing names, verifies, or diffs a whole *version* of the module. The Merkle DAG solves exactly this. It stores no meaning by itself; it maintains references by hash and produces a single root that commits to the exact set and structure of blocks in a version. That root *is* the identity of the version, and it is what makes Principle 3 (identifiable, verifiable, and reconstructible versions) concrete.

**How it works.** The leaves of the DAG are block\_ids (content hashes); each internal node hashes the hashes of its children; the root hashes down to everything below it. Changing any block changes its hash, which changes every ancestor up to the root. Three operations follow directly. *Integrity* is checked by recomputing hashes from the leaves up and comparing the root. *Membership* of a single block is proven with the sibling hashes along its path to the root (a Merkle proof of size  $O(\log n)$ ), without downloading the rest of the module. *Differencing* two versions compares roots top-down and descends only where child hashes differ, so the cost is proportional to what changed, not to the size of the module.

**Why a DAG rather than a tree.** Because blocks are shared across versions. When a new version reuses unchanged blocks, those blocks are shared by hash instead of duplicated, so a node may be referenced by more than one root; the structure is therefore a directed acyclic graph, as in Git’s object store. This structural sharing is what makes incremental updates cheap (Section 7): a new version adds only the changed blocks and a new root, and reuses everything else by reference. What sharing guarantees is precisely this reuse of blocks, which are the bytes a consumer would otherwise have to transfer; how much of the DAG’s internal structure above the leaves is also reused depends on the layout a conforming implementation chooses, and internal nodes carry no content of their own, being hashes of hashes.

### Example

semantic-root v1 contains K1, K2, and K3. If K2 is corrected, a new K2' is created and semantic-root v2 references K1, K2', and K3. K2 is never modified in place, and K1 and K3 are shared between the two roots (Figure 4).

**Two Merkle DAGs.** It is worth separating the *internal* Merkle DAG described here, which versions the blocks inside a module, from the *external* Merkle DAG that OCI itself forms over manifests, descriptors, and blobs to version whole modules (Section 7, Figure 5). The same idea is applied at two levels: blocks within a module, and modules within a brain.

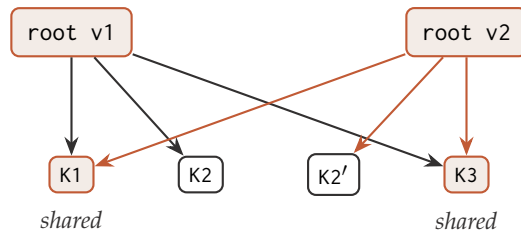


Figure 4: Structural sharing in the internal Merkle DAG. Correcting K2 into K2' yields a new root v2 that reuses K1 and K3 by hash; only the changed block and the new root are added. Solid dark edges belong to v1, accent edges to v2.

**Storage-independent identity.** Because the root depends only on the blocks and their structure, it is a canonical identity of a knowledge state, independent of how that state is stored or transported. Two parties that assembled the same blocks obtain the same root and can verify they hold identical knowledge; the OCI digest, by contrast, identifies a particular physical file or manifest.

### 6.3 Indices

Indices are derived: they can be rebuilt from the blocks. They may travel inside the same module so that each consumer need not recompute them, but they never replace the content. Table 5 lists the indices and their roles.

Table 5: Indices, their typical queries, and their role.

| Index               | Typical query                               | Role                           |
|---------------------|---------------------------------------------|--------------------------------|
| Hash map            | Get exactly sha256:ABC.                     | Immediate physical resolution. |
| B <sup>+</sup> tree | Episodes between two dates; concept by ID.  | Ordered keys and range scans.  |
| Inverted            | Blocks containing “Fourier”.                | Lexical matching.              |
| Vector              | “Decompose a periodic function into sines”. | Semantic similarity.           |
| Graph               | Related concepts, evidence, dependencies.   | Associative traversal.         |
| Bitmap / facets     | semantic AND subject=signals AND current.   | Efficient filter intersection. |

Whether an index travels inside the module or is rebuilt on installation depends on whether it can be regenerated without a model. Structural indices (hash map, B<sup>+</sup> tree, inverted, bitmap, and, when relations live on the blocks, the graph) are deterministic functions of the blocks and can be rebuilt cheaply by any client. The vector index is the exception: rebuilding it requires an embedding model, which a model-agnostic client does not carry. The vector index therefore travels with the module and records the embedding model and version that produced it, so that consumers can reproduce or compare it. This keeps the blocks as the single source of truth while acknowledging that one derived structure cannot be regenerated in isolation.

The same argument applies to a client that restarts. A structural index can be rebuilt from the installed blocks whenever it is needed, so losing it costs only the time to recompute it; the vector index cannot, and a client that holds it only in memory has lost it for good. An implementation must therefore persist the travelling index locally, alongside the module it belongs to, and must not republish a placeholder in its stead: an artifact whose layer claims a vector index and carries

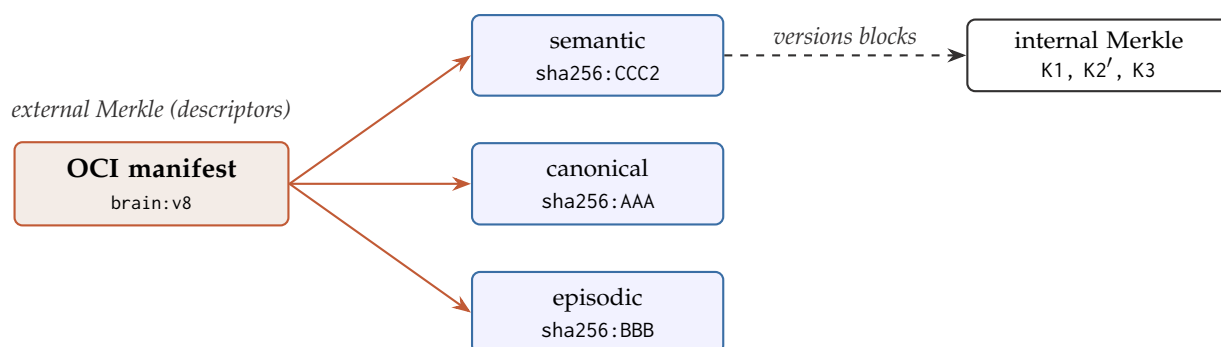


Figure 5: OCI distributes modules as separate blobs referenced by a small manifest; inside each module, an internal Merkle DAG versions the individual knowledge blocks.

none is worse than one that omits the layer, because a consumer can detect the absence and cannot detect the emptiness.

This separation has a representational reading. A block carries meaning in two portable, *symbolic* forms: its text and its explicit relations to other blocks, the edges that in aggregate form the knowledge graph. Learned, *sub-symbolic* representations (embeddings) are never stored in the block; they live in the vector index as derived, model-tagged views. Operating mathematically on concepts, such as vector arithmetic or steering, happens inside the external LLM in its own representation space, not in the brain. The brain thus owns durable symbolic knowledge and borrows the operational vector space from whichever model queries it. This also clarifies a naming coincidence: the *semantic* module denotes general, consolidated knowledge (as opposed to episodic memory), not an embedding-based representation.

## 6.4 Three levels of hashes

A recurring source of confusion is that the same cryptographic algorithm is used for three different kinds of identity. Table 6 separates them.

Table 6: Three levels of hashes.

| Level       | What it identifies                       | Example                 |
|-------------|------------------------------------------|-------------------------|
| Block hash  | A logical unit of knowledge.             | sha256:FOURIER_CONCEPT  |
| Merkle root | The logical composition of a module.     | sha256:SEMANTIC_ROOT_V7 |
| OCI digest  | The published physical blob or manifest. | sha256:OCI_MODULE_BLOB  |

These values may use the same algorithm, but they do not represent the same thing. The block hash is *knowledge identity*; the Merkle root is the identity of a *logical snapshot*; the OCI digest is the identity of a transportable *file or manifest*.

## 7 Distribution with OCI Artifacts

OCI is a standard for content-addressed manifests, descriptors, and blobs. A compatible registry stores the blobs and allows referencing them by tag or digest; the specification organizes its components as an external Merkle DAG of descriptors [16–18]. Figure 5 shows how OCI distributes modules while the internal Merkle versions the blocks within each module.

### 7.1 Complete brain

A brain can be published as an OCI Artifact that references one blob per module:

```

monotributo-brain:v3
  canonical -> sha256:AAA
  episodic  -> sha256:BBB
  semantic   -> sha256:CCC
  procedural -> sha256:DDD
  provenance -> sha256:EEE

```

The OCI manifest is small. Downloading the manifest does not imply downloading all modules.

## 7.2 Selective installation

To install only the episodic module, the client resolves the descriptor marked as episodic and downloads sha256:BBB. It does not need to download canonical, semantic, or procedural. It then opens the episodic Merkle DAG and its local indices.

### Necessary condition

Selection is only efficient if each module is a separate OCI blob or artifact. If everything is mixed into a single file, the whole file must be downloaded.

## 7.3 Incremental update

If only the semantic module changes, the new manifest can reuse the digests of the remaining modules. The consumer downloads only the new semantic blob; inside it, the Merkle root reveals which logical blocks changed.

Table 7: An incremental update touches only the changed module's digest.

| Version | Canonical | Episodic | Semantic | Procedural |
|---------|-----------|----------|----------|------------|
| v7      | AAA       | BBB      | CCC      | DDD        |
| v8      | AAA       | BBB      | CCC2     | DDD        |

OCI contributes storage, registry authentication, transport, digest-based deduplication, and tags. Boltzmann Brain contributes the meaning of the modules, the blocks inside them, and their query contract.

## 8 The Boltzmann Protocol

An installed OCI Artifact is passive data. The *Boltzmann Protocol* is the contract through which any conforming client interacts with a brain, along two paths, query and ingestion. Boltzmann Brain is therefore delivered as a protocol rather than as a deployable service or container image: the brain is portable data, and any client that speaks the protocol can read and extend it. The protocol includes no LLM of its own; it governs structure, identity, validation, indices, snapshots, and distribution. Concretely, it defines operations to:

- Discover installed modules, versions, and Merkle roots.
- Open B<sup>+</sup>, lexical, vector, and graph indices.
- Resolve a block\_id to its physical location and verify its hash.
- Register, replace, and drop canonical sources; record provenance and derived\_from relations.

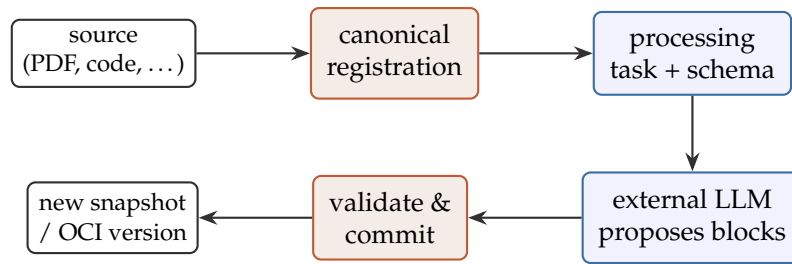


Figure 6: The brain conserves and validates; the external LLM interprets and proposes blocks. Only validated candidates are committed and versioned.

- Define processing tasks and output schemas for an external LLM.
- Validate candidate blocks before they are incorporated into the brain.
- Update Merkle DAGs, indices, and snapshots.
- Drop, supersede, demote, collect, and redact blocks under an explicit policy; a drop rebuilds the affected Merkle DAGs and cascades through provenance (Section 11).
- Publish new versions as OCI Artifacts.
- Be exposed through a skill, MCP, CLI, or API.

Because these are protocol operations rather than a bundled engine, they can be implemented independently and in any language, and the brain never depends on a particular process being kept running.

## 8.1 The boundary between the protocol and the external LLM

The protocol defines which memory types exist and which schema each block must satisfy. The external LLM performs the semantic operations: reading, interpreting, extracting concepts, recognizing episodes, proposing procedures, and contextualizing results. *The LLM proposes; the protocol governs what is stored.*

This boundary keeps the brain provider-independent. The same snapshot can be used with a commercial service, a development agent, or a local model, as long as the client implements the Boltzmann Protocol. Trust does not rest on a privileged service: because every block, snapshot, and module is addressed by its hash, any client can verify integrity independently of who produced it.

### Design rule

The LLM never writes directly to the Merkle DAGs or to the indices.

## 9 Ingestion via an External LLM

Ingestion separates the preservation of a source from its interpretation (Figure 6). First the original is registered deterministically; then the protocol delegates semantic processing to the user's LLM.

### 9.1 Canonical registration

Canonical evidence enters through three protocol operations. There is no in-place edit of a source: a new edition is always a new block.



**Register.**

1. The user provides a PDF, text, repository, conversation, image, or other source.
2. The LLM invokes the `brain.ingest` operation through the available skill or adapter.
3. The format is validated, its SHA-256 computed, and exact duplicates detected; re-registering an identical blob is a no-op.
4. The original is stored as an immutable canonical block, optionally together with a normalized view produced by a named deterministic pipeline.
5. The protocol records who incorporated it, when, from where, and under what license or policy.
6. The canonical Merkle root changes and a new snapshot becomes available, even though no semantic interpretation exists yet.

Registering a source does not declare it true. The canonical module asserts that the evidence was incorporated and preserved.

**Replace.** When a newer edition of a source should take precedence, the protocol registers the new original as a distinct block and records a supersedes relation to the previous one. That relation is a provenance edge rather than a field of either block, so both remain pure statements about the bytes they preserve and the precedence between them stays auditable where every other actor-dependent record lives. The caller chooses whether the superseded original remains in the current composition (kept for audit) or is dropped in the same commit. In either case the new block is what subsequent processing tasks read. `replace` is therefore `register` plus a supersession edge, with an optional drop of the old evidence, not a mutation of bytes already stored.

**Drop.** Excluding a canonical block from the current composition is the same drop operation defined in Section 11, with a privileged cascade: every derived block that cites the dropped evidence is dropped by default in the same commit, unless the caller has already registered a replacement and requests re-derivation against it. Dropping canonical evidence is allowed only under explicit policy (Principle 2).

**9.2 Processing task**

After canonical registration, the protocol returns a structured task to the LLM. The task identifies the source, the instructions, the allowed memory types, the required references, and the exact output schema:

```
{
  "operation": "extract_knowledge",
  "source": "sha256:PDF123",
  "allowed_memory_types": ["episodic", "semantic", "procedural"],
  "requirements": ["cite source ranges", "do not invent"],
  "output_schema": "boltzmann.candidates/v1"
}
```

The LLM reads the source and returns typed candidate blocks. A document may produce semantic and procedural memory without producing episodic memory; the classification depends on what the source actually contains.

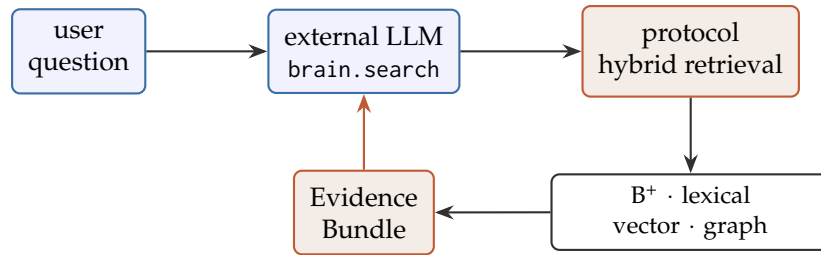


Figure 7: The external LLM queries the brain and contextualizes the returned Evidence Bundle. The protocol combines indices, verifies hashes, and returns data rather than prose.

### 9.3 Validation and commit

Results from the LLM do not enter automatically. The protocol checks schema, references, pages, types, duplicates, relations, and basic contradictions. Candidates may end up `VALIDATED`, `PENDING_REVIEW`, `REJECTED`, or `CONTRADICTED`. For each accepted block, the protocol prescribes the following steps:

1. Serializes the block canonically.
2. Computes its `block_id` by hashing.
3. Stores the block as an immutable object.
4. Connects it to the source via provenance.
5. Incorporates it into the episodic, semantic, or procedural Merkle DAG.
6. Updates  $B^+$  trees, lexical, vector, and graph indices as appropriate.
7. Creates a new snapshot and, when published, a new OCI version.

## 10 Query via an External LLM

For now, Boltzmann Brain is conceived as a medium for LLMs. The user converses with their usual LLM; a skill or adapter lets that LLM query the brain through the protocol. The brain returns raw or semi-raw knowledge, and the LLM turns it into an answer appropriate for the task (Figure 7).

### 10.1 Query flow

1. The user asks the external LLM a question.
2. The LLM interprets the intent and calls `brain.search` with the query and optional filters.
3. The protocol combines  $B^+$  trees, lexical search, vectors, and graph.
4. Knowledge blocks and their canonical sources are retrieved.
5. The protocol verifies hashes and membership in the installed snapshot.
6. An Evidence Bundle is returned.
7. The LLM explains, summarizes, compares, or uses the evidence.

## 10.2 Query planning and guarantees

Step three above hides a query planner. By Principle 7 a query is declarative: the caller expresses intent (a query, optional filters over memory type, subject, or recency, and optional hints such as a retrieval mode or a result limit) and never names a physical index. Choosing which indices to use, and how to combine them, is the responsibility of a conforming implementation, in the same way that the SQL standard defines a query language and its semantics while each database ships its own optimizer. Selection follows the shape of the query: exact identifiers resolve through the hash map, ordered or range predicates through the B<sup>+</sup> tree, term matches through the inverted index, natural-language intent through the vector index, neighborhood and traversal through the graph, and categorical filters through bitmaps. In practice retrieval is hybrid: filters narrow a candidate set, lexical and vector search generate candidates in parallel, their rankings are fused, the graph expands from the strongest hits, and the hash map resolves and verifies each result.

The protocol fixes the contract and the invariants, not the algorithm. A conforming implementation must return knowledge blocks with their provenance and a retrieval score, never prose; every returned block must be verified against the installed snapshot by hash and membership; and no single index may be treated as authoritative. The fusion method, ranking weights, graph expansion depth, and any adaptive intent classification are deliberately left to the implementation. A consequence worth stating explicitly is that the protocol guarantees *verifiability, not identical ranking*: two conforming implementations may return the same verifiable set of blocks in a different order, because vector search is approximate and ranking is tunable.

## 10.3 Evidence Bundle

The brain's result is not a written answer. It is a data contract that may include matching blocks, content, memory type, relations, sources, retrieval score, block IDs, and verification status:

```
{
  "matches": [{
    "block_id": "sha256:CONCEPT1",
    "memory_type": "semantic",
    "content": "<serialized knowledge block content>",
    "source": {"block_id": "sha256:PDF123", "page": 147},
    "verified": true
  }]
}
```

The consumer receives block identities, content, memory type, sources, and verification status, and remains free to explain, summarize, compare, or reuse the evidence. Those transformations happen outside the brain.

# 11 Retention, Forgetting, and Erasure

Everything described so far only adds. Content addressing and immutable blocks make individual units append-only, but a module version is a *composition* of those units, and compositions can exclude what should no longer belong. Without an explicit way to drop blocks, a brain that only grows is neither practical nor, in some jurisdictions, lawful, and it cannot recover from knowledge that was deliberately wrong.

## 11.1 Why exclusion must be first-class

Three pressures call for different responses, and they should not be conflated.

**Wrong or obsolete knowledge.** A semantic definition may be incorrect, a procedure may be outdated, or a canonical source may have been ingested in error. Leaving the block in the current composition and merely down-ranking it is not enough: the wrong claim still lives in the snapshot that consumers install. The protocol needs a way to *exclude* it from the next version.

**Cost and signal.** Unbounded growth costs storage and precision. Stale near-duplicates compete for the top of every ranking, so a brain that never excludes gets measurably worse at answering.

**Law and safety.** Personal data, credentials, or licensed material may have to disappear even from retained history [7]. That case is stricter than exclusion: it requires destroying bytes that a retained root still names. It is treated separately in Section 11.5.

## 11.2 What biology suggests

Forgetting in nervous systems is a regulated process, not a storage failure. Microglial pruning eliminates connections that are not reinforced [19], active forgetting is driven by dedicated molecular machinery [6], and engram studies treat inaccessibility as adaptive plasticity rather than erasure [22]. Cognitive psychology separates the availability of a trace from its accessibility [2]. The architectural reading is that exclusion from the current composition is the default, that reclamation of what no retained version needs is a background process, and that destroying evidence is a rare, policy-bound exception rather than the ordinary cleanup path.

## 11.3 Drop: excluding blocks from a module

The primary removal operation is *drop*. It applies to the **canonical**, **semantic**, **procedural**, and **provenance** modules. The **episodic** module is exempt: episodes are a chronological record of what happened, so corrections append new episodes or supersession relations rather than rewriting the past (Section 5).

A drop does not mutate blocks in place. Blocks remain content-addressed and immutable. What changes is the *composition* of the module: the protocol builds a new Merkle DAG whose leaves are the surviving `block_ids`, computes a new root, rebuilds the module's indices from that composition, and records the removal in provenance (Figure 8). The dropped blocks become unreachable from the new root and are eligible for pruning once no retained root references them. Structural sharing still applies: unchanged blocks are reused by hash, so the cost of a drop is proportional to what was removed, not to the size of the module (Section 6).

### Example

semantic-root v1 contains K1, K2, and K3, where K2 is a wrong Fourier definition. A drop of K2 yields semantic-root v2 over K1 and K3 only. Indices are rebuilt without K2, and provenance records that K2 was dropped, by whom, and why. Optionally the external LLM re-derives a corrected K2' from the canonical source and a later commit adds it under v3. The wrong block never appears in v2 or v3.

**Provenance cascade.** Dropping a block is not always local, and the cascade differs by module.

Dropping a *semantic* or *procedural* block removes it from that module's composition and rewrites or removes provenance edges that pointed to it. Downstream dependents are treated the same way. The canonical source remains, so the caller may later re-derive a corrected block from the same evidence.

Dropping a *canonical* block is privileged. Because derived knowledge cites that evidence as its root, the protocol always walks the dependency closure and, by default, *drops* every semantic and procedural block that listed the canonical as evidence, in the same commit. Re-derivation is not the default: it runs only if the caller has already registered a replacement canonical (*replace*) or explicitly requests *rederive* against another canonical still in the composition. Each affected module receives a new Merkle root, so a single logical removal of evidence can publish several new module versions at once. Provenance records the actor, the policy, the reason, and the resulting roots, so the removal stays auditable even though the dropped content is no longer part of the current composition.

#### Example

The wrong lecture PDF was ingested as `PDF_wrong`, and several semantic blocks cite it. A drop of `PDF_wrong` yields a new canonical root without that blob; the cascade drops the derived definitions in the same commit. Registering the correct PDF and running a processing task then produces new candidates under a later snapshot. The wrong source never appears in the current composition, and neither do interpretations that depended only on it.

**Batch invalidation.** Because provenance records the producer of each derived block (model, pipeline, ingestion batch, or actor), a drop may be stated over a *set*: everything produced by a given ingestion, or everything derived by a given model version. That is the natural response to deliberately wrong knowledge introduced in bulk, and it reuses the same cascade rather than inventing a second mechanism.

## 11.4 Companion mechanisms

Drop is the ordinary way to remove knowledge from a non-episodic module. Three companion operations cover accessibility, reclamation, and the rare case where bytes must disappear while a retained root still names them (Table 8).

**Supersession and demotion.** A block is marked as superseded by a newer one, or its retrieval priority decays. It stays in the current composition and remains verifiable; what changes is accessibility. This is the default path for the episodic module, and an optional soft path elsewhere when history inside the current snapshot must remain queryable under an explicit filter.

**Pruning of unreachable blocks.** A brain retains a set of roots (tagged releases plus recent snapshots). After drops, former leaves are unreachable from those roots and can be reclaimed by mark-and-sweep [3]. Pruning never decides *what* to forget; it reclaims what no retained composition still needs.

**Redaction of referenced content.** Only when content that a retained root still references must be destroyed does the protocol remove bytes: a *tombstone* deletes the bytes and leaves an auditable record; *crypto-shredding* destroys the encryption key [10]; a *lineage rewrite* rebuilds history so the block was never referenced. Redaction is the path for destroying retained canonical bytes under law or safety policy. It is not the cleanup path for a wrongly ingested source or for wrong semantic knowledge: those are dropped.

Table 8: Removal mechanisms and their guarantees.

| Mechanism                                               | What it removes                       | Reversible                       | Effect on current roots                                                                                               |
|---------------------------------------------------------|---------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Drop (non-episodic)                                     | Membership in the module composition. | Via a later commit that re-adds. | New Merkle root without the block; indices rebuilt; provenance updated. Canonical drops always cascade to dependents. |
| Supersession / demotion                                 | Accessibility, not membership.        | Yes                              | None; block stays in the composition.                                                                                 |
| Pruning                                                 | Bytes of blocks in no retained root.  | No                               | None; retained roots unaffected.                                                                                      |
| <i>Redaction (retained root still names the block):</i> |                                       |                                  |                                                                                                                       |
| Tombstone                                               | The bytes; identity kept.             | No                               | Root intact; block unresolvable.                                                                                      |
| Crypto-shredding                                        | The plaintext, via the key.           | No                               | Root intact; ciphertext still verifies.                                                                               |
| Lineage rewrite                                         | The block and its history.            | No                               | Prior roots invalidated; new lineage published.                                                                       |

## 11.5 Why drop preserves a clean snapshot

A drop publishes a new composition. Consumers of the new root never see the dropped block: membership proofs, indices, and Evidence Bundles are computed over the survivors only. Older roots, if still retained, continue to verify exactly as before, because nothing about them changed. The brain therefore offers a *clean current snapshot* after every drop, which is the property that makes exclusion usable for wrong knowledge.

Redaction is different: it punches a hole in a composition that still names the block. The Merkle DAG references identities, not bytes, so deleting the bytes behind a leaf changes no hash and membership still verifies, but reconstruction of that one block is forfeited. A conforming implementation must report which blocks of a snapshot are resolvable and which are tombstoned, so that a removed block is never indistinguishable from a corrupted one. Provenance is again the home for that record.

Two limitations of redaction deserve stating. A hash of low-entropy content is not anonymous, so confirming a guess may still be possible when the `block_id` is kept. And erasure does not propagate by itself across already-pulled copies: the protocol can publish a revocation, but

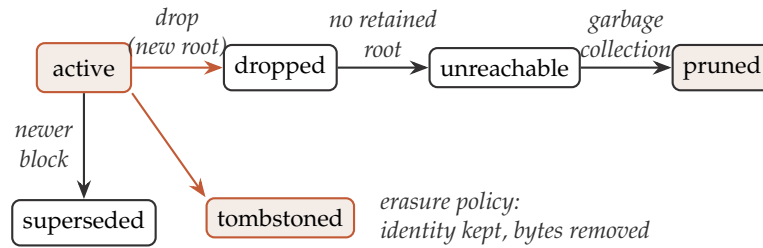


Figure 8: Lifecycle states of a knowledge block. The ordinary removal path for non-episodic modules is drop: a new Merkle root excludes the block, provenance is updated, and pruning later reclaims bytes once no retained root references it. Supersession keeps the block in the composition; redaction destroys bytes that a retained root still names.

a distributed brain can only signal destruction, not guarantee it. Neither limitation applies to an ordinary drop of wrong semantic or procedural knowledge, because the new root simply never included the block.

## 11.6 Policy is configuration

As with query planning (Section 10), the protocol fixes the operations and the invariants, not the thresholds. Who may drop from which module, how deep the provenance cascade runs before requiring review, how many roots to retain, and which categories of content are redactable are deployment decisions. What the protocol requires is that every drop, supersession, prune, and redaction be explicit, recorded in provenance, and reportable, which is the content of Principle 8.

## 12 Knowledge Lifecycle

The full lifecycle ties the previous sections together:

1. **Source:** the user provides information via their LLM and a skill.
2. **Canonical registration:** the protocol preserves the original, computes its hash, and records provenance.
3. **Delegation:** the protocol returns a task and a processing schema.
4. **Interpretation:** the LLM proposes episodic, semantic, and procedural blocks and relations.
5. **Validation:** the protocol checks structure, references, duplicates, and policies.
6. **Commit:** accepted candidates receive hashes and are stored as immutable blocks.
7. **Indexing:** B<sup>+</sup> trees, lexical search, vectors, and graph are updated.
8. **Versioning:** the Merkle roots of the affected modules change.
9. **Publication:** OCI distributes only the modified physical modules.
10. **Query:** the LLM obtains Evidence Bundles and transforms them for the user.
11. **Reconsolidation:** new evidence or corrections generate later candidates and snapshots.
12. **Forgetting and retention:** non-episodic modules drop wrong or obsolete blocks, rebuild their Merkle roots, and cascade through provenance (canonical drops always cascade); prune what no retained root needs; episodic memory supersedes; redaction destroys retained bytes only when policy requires it (Section 11).



The canonical, episodic, semantic, procedural, and provenance memories remain at the center of the architecture. The change consists solely in decoupling cognitive processing: the protocol organizes and governs; the external LLM interprets and contextualizes.

## 13 Conclusion and Future Work

We presented Boltzmann Brain, a conceptual architecture for portable, verifiable, and model-agnostic knowledge. Its core commitment is a clean separation of concerns: the brain conserves, validates, and retrieves, while an external, interchangeable LLM interprets and contextualizes. From this commitment follow the five-module memory model, the content-addressed blocks versioned by Merkle DAGs, the derived indices, the three levels of hashes, a distribution model in which OCI Artifacts enable selective installation and incremental updates, and an account of how canonical evidence roots re-derivation, how non-episodic modules drop wrong knowledge with a privileged cascade when that knowledge's source is excluded, rebuild Merkle roots, and reclaim unreachable blocks, with redaction reserved for destroying retained bytes.

This version deliberately fixes boundaries and responsibilities rather than implementations. Concrete block schemas, wire formats, and algorithms remain to be validated: the exact block schema per memory type, the deterministic serialization used for `block_id`, the consolidation and contradiction-detection algorithms, the pack sizing and index engine choices, and the precise shape of the ingestion and query operations exposed through skills, MCP, CLI, or API. Retention leaves its own open questions: the exact shape of the drop operation and its provenance cascade, which decay function governs demotion in the episodic module, what a sensible default retention policy over roots looks like, and how a revocation should propagate through registries and caches so that destruction of retained bytes becomes more than a local guarantee. Validating these pieces against real, multi-source brains is the natural next step.

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